WCET Analysis Methods
Milestone 2: Scientific Context and Tradeoffs

Muhammad Umer Hayat
muhammad-umer.hayat@stud.hshl.de

1 Goal of Milestone 2

- Place WCET analysis in the larger real-time and embedded systems context.
- Identify related approaches and neighboring problem classes.
- Compare method families by their engineering tradeoffs.
- Clarify what the methods mean for realistic embedded systems.

2 Where WCET Fits

- **Real-time scheduling:** scheduling tests need execution-time bounds as input.
  - Example: Rate Monotonic (RM), Earliest Deadline First (EDF), and response-time analysis.
  - Basic idea: if the WCET value is wrong, the scheduling proof becomes weak or invalid.
- **Formal methods:** Static WCET estimation is based on over-approximating paths and machine state formally.
- **Computer architecture:** Caches, pipelines, branch prediction, prefetching, and inter-core shared resources result in timing ambiguity.
- **Safety certification:** Avionics and automotive systems need to prove that deadlines are not just possible but justified.

3 Context Map

Area	Connection to WCET	Why it matters
Real-time scheduling	WCET is used as task execution-time parameter C_i.	Wrong C_i can make a schedulability result meaningless.
Formal verification	Static methods try to prove safe upper bounds.	Gives strong guarantees, but depends on correct models.
Processor architecture	Hardware behavior influences execution time.	Modern processors are fast but harder to predict.
Safety standards	Standards decide what evidence is acceptable.	A method can be technically interesting but still not certifiable.

Tool support	Tools implement static, measurement-based, or hybrid workflows.	Practical use depends on automation, cost, and available hardware details.

4 Related and Neighboring Approaches

Approach / concept	**Relation to WCET**	**Difference or limitation**
Worst-Case Response Time (WCRT)	Extends from one task to a task set. Uses WCET as input.	Includes preemption, blocking, and interference between tasks.
Abstract interpretation	Mathematical basis of many static WCET tools.	Sound but may become pessimistic and model-heavy.
Time-triggered architectures	Reduce timing uncertainty by fixed schedules or time slots.	Simpler timing analysis, but less flexible and sometimes lower average performance.
Resource isolation / partitioning	Limits interference from caches, buses, or memory controllers.	Helps WCET analysis but can waste hardware capacity.
Extreme Value Theory (EVT)	Used in probabilistic timing analysis to estimate rare execution-time tails.	Requires statistical assumptions such as independence and representativeness.
Runtime monitoring	Observes timing behavior during execution.	Detects problems late; does not replace offline WCET evidence.
Machine learning prediction	Can estimate execution time trends from data.	No strong safety guarantee, so weak for certification.

5 Main WCET Method Families and Tradeoffs

Method	**Useful because**	**Main cost / weakness**	**Trust note**
SDTA	Gives a deterministic upper bound without running every input.	Needs a correct hardware timing model and flow facts.	Strong when hardware is simple and documented.
MBDTA	Practical because it measures on real hardware.	The true worst case may not appear during testing.	Easy to apply, but trust depends heavily on test quality.
HYDTA	Combines measurements with static path information.	Separate measurements may not preserve cache or memory state.	Better coverage than pure measurement, but still not fully safe for complex platforms.
SPTA	Uses probability with static reasoning.	Usually limited to simple or randomized timing models.	Theoretically interesting, but narrow practical use.
MBPTA	Uses measurements and statistics to estimate probabilistic WCET.	Needs time-randomized behavior and strong statistical conditions.	Promising for complex hardware, but certification is still difficult.

HYPTA	Reduces measurement effort by combining probabilistic and structural information.	Validity can be unclear if analysis uses a modified or abstracted program.	Useful research direction, but less mature.

6 Engineering Tradeoffs

- **Formal guarantee vs practical deployability**
  - SDTA gives the strongest kind of argument.
  - MBDTA is easier to use, but does not prove that the worst case was tested.
- **Precision vs analysis cost**
  - More detailed hardware models can reduce pessimism.
  - But building and validating those models increases effort and cost.
- **Single-core vs multicore**
  - Most classical WCET assumptions are easier on single-core systems.
  - Multicore systems create interference through shared caches, buses, and memory controllers.
- **Offline analysis vs online behavior**
  - WCET is usually computed offline before deployment.
  - Adaptive systems may change behavior at runtime, which weakens old assumptions.
- **Certification evidence vs research potential**
  - MBPTA is promising for modern processors.
  - But production hardware and certification arguments are not mature enough yet.

7 Consequences for Embedded Systems

System type	**Suitable direction**	**Reason**
Simple microcontroller ECU	SDTA	Simple in-order hardware is easier to model.
Long-used single-core railway system	MBDTA or HYDTA	Platform behavior is stable and test campaigns are manageable.
Mixed-criticality multicore platform	MBPTA, with caution	Different tasks may need different confidence levels. Certification remains open.
Autonomous vehicle / UAV software	No fully satisfying method today	Large input space plus multicore interference makes trust difficult.
FPGA / GPU-based acceleration	Still underdeveloped WCET area	Parallelism and reconfigurability break many CPU-based assumptions.

8 Why Some Related Work Is Sparse

- Industrial hardware details are often confidential or incomplete.
- Timing behavior depends on low-level processor features that are hard to publish and reproduce.
- Certification practice changes slower than research papers.
- GPU, FPGA, and multicore WCET are harder than classical single-core CPU analysis.
- Many modern software workloads, especially AI workloads, do not fit old assumptions about bounded loops and predictable memory access.

9 Feedback Questions

- Is MBPTA currently the strongest research direction for multicore WCET, or is resource isolation more realistic?
- For a seminar paper, should the focus be mainly on trustworthiness, certifiability, or multicore scalability?

References

[1] R. Wilhelm et al., "The worst-case execution time problem: Overview of methods and survey of tools," *ACM Transactions on Embedded Computing Systems*, vol. 7, no. 3, pp. 1-53, 2008.

[2] L. Cucu-Grosjean et al., "Measurement-based probabilistic timing analysis for multi-path programs," in *Proc. Euromicro Conference on Real-Time Systems (ECRTS)*, 2012.